

# Component-Dependent Sensitivity to Stale Predictions in ACT Action-Chunk Execution

**Abstract**—How old can a predicted action be before it should be refreshed? Action-chunk policies forecast a trajectory from each observation, yet execution rules apply one temporal setting to the action vector. We diagnose component dependence by taking arm and gripper commands from ACT chunks while aligning both to the same physical control step. These probes query the policy once per step and are diagnostic interventions, not deployment schedules. In a preregistered 140-block ACT evaluation, moving a one-second-old prediction from the gripper to the arm changed success from 83/140 to 38/140, a 32.14-point difference. On a separate development cohort, the fresh-arm/stale-gripper branch remained above the stale-arm/fresh-gripper branch from 0.10 to 1.60 s, with observed separations of 21.43–54.76 points. Translation staleness was also substantially more damaging than rotation staleness. In executable schedules, extending gripper commitment improved success by 4.667 and 5.778 points at two arm cadences with nearly matched policy-query rates. Coherent H16 nevertheless remained the strongest operating point. Same-target disagreement ranked the arm components in the opposite order from behavior, and the probe couples source age with prediction lookahead. The mechanism and generality beyond the evaluated ACT policies therefore remain unresolved.

## I. Introduction

An action-chunk policy predicts commands for several future control steps from each observation [1]. Its executor must then decide how long or how repeatedly to use those predictions before requesting a new chunk. Common executors expose one temporal setting for the full action vector, although the vector combines translation, rotation, and gripper commands. A single setting is behaviorally neutral only if these components respond similarly when their predictions become stale.

The evaluated ACT policy shows a striking component structure on the Object development cohort. At a source age of 1.00 s, success was 8.73% with stale translation, 9.52% with the full arm stale, 42.06% with stale rotation, 44.44% with all components fresh, and 64.29% with a stale gripper. These exactly block-matched values combine audited historical anchors with later reviewer-directed component probes; they are a comparative characterization, not one preregistered five-condition factorial. Staling translation alone nearly reproduces the degradation seen when the full arm is stale. Rotation staleness incurred little detectable cost relative to Fresh on this cohort (−2.38 points,  $p = 0.607$ , with both intervals including zero). The magnitude of the stale-gripper simple effect varies substantially across cohorts, so the 64.29% value is not evidence of a general benefit.

We measure this structure with a same-target diagnostic. At control step  $t$ , one component can use offset 0 from the chunk queried at  $t$ , while another can use offset 20 from the chunk queried 20 steps earlier. Both entries were predicted for physical step  $t$ . More generally, query time  $q$  and chunk offset  $k$  obey  $q + k = t$ . Holding the target fixed therefore couples an older source observation to a farther-ahead chunk prediction. We measure their joint effect and do not separate source age from prediction lookahead. Every probe queries the policy once per executed step, making it a diagnostic intervention rather than a deployment executor.

This study makes three contributions. First, a pre-registered same-target test shows a large arm–grripper asymmetry, and a separate development sweep preserves its ordering over source ages from 0.10 to 1.60 s. Second, component probes expose strong translation–rotation heterogeneity within the arm. The most natural same-target disagreement metric ranks these components in the opposite order from their behavioral sensitivity. Third, executable component-resolved schedules improve success when gripper commitment is extended at two fixed arm replanning cadences, while coherent H16 remains the strongest overall operating point.

The claims are deliberately bounded. The same-target results diagnose temporal source assignment under a policy-query rate of 1, whereas deployment schedules replan less often. The central behavioral evidence concerns the evaluated ACT policies in LIBERO [2]; it does not establish cross-policy generality. Coherent H16 remains strongest in the frozen static benchmark, and the diagnostics tested here do not identify the mechanism behind the component ordering.

## II. Related Work

### A. Action chunking and temporal ensembling

ACT predicts action chunks to shorten the effective decision horizon and can aggregate overlapping same-target predictions at every control step [1]. Diffusion Policy combines a receding-horizon controller with a diffusion model that generates an action sequence from recent observations [3]. Both establish chunk execution as part of the policy interface rather than a neutral implementation detail. Recent analyses further show how action chunks encode temporal dependencies and how policy context changes the value of open-loop commitment [4], [5]. Our experiments keep the trained

chunk policy fixed and manipulate only the temporal source used at execution.

### B. Whole-policy execution adaptation

Adaptive executors decide how much of a predicted chunk to commit before replanning. BID samples candidate chunks and selects among them using backward coherence with earlier decisions and forward contrast among future plans [6]. AutoHorizon uses model attention to estimate a shared predictive limit for each chunk [7]. PACE selects a shared horizon from low-speed valleys in the predicted motion [8]. DEHP learns a lightweight horizon predictor with reinforcement learning while freezing the base chunk policy [9]. BCP learns a sequence of continue-or-replan decisions over a chunk [10]. VLA-Corrector monitors visual-feature deviation and truncates a chunk when corrective replanning is indicated [11]. A3 instead accepts the longest action prefix that passes its verification procedure [12]. Adaptive Action Chunking uses action entropy from the current prediction to select its inference-time chunk size [13]. These signals and objectives differ, but each method makes a temporal decision for the predicted action as a whole.

### C. Stale predictions and asynchronous execution

Staleness also arises when inference and control run on different clocks. RTC addresses asynchronous chunk generation by preserving actions that must execute and inpainting a consistent continuation of the next chunk [14]. Same-target alignment asks a complementary measurement question: how does source age affect selected components when every command still targets its intended physical step? The diagnostic does not isolate latency or computational cost, and it couples observation age with prediction lookahead.

### D. Heterogeneous action generation

Robot actions can also be structured at the representation and generation levels. ARP permits different chunk sizes for different action types [15]. Mixture of Horizons processes several prediction horizons with a shared action transformer, fuses them through a learned gate, and can select executable prefixes from cross-horizon consensus [16]. Both alter the policy or action-generation structure. Our policy remains fixed and predicts one seven-dimensional chunk; component identity enters only when the executor selects which same-target source supplies translation, rotation, or gripper commands.

We do not propose an alternative adaptive-horizon executor. We ask whether a temporal decision applied to an action chunk is behaviorally uniform across its action components, and we measure that dependence rather than optimize a new executor. Track A tests a bounded operational consequence at fixed arm cadences; it is not a claim of superiority over adaptive methods.

## III. Same-Target Temporal Probe

At control step  $t$ , a Fresh component uses offset 0 from the chunk predicted from the current observation. A component with source age  $d$  instead uses offset  $d$  from the chunk queried  $d$  steps earlier. Both entries were originally predicted for the same physical step  $t$ . For example, A0G20 takes the arm command from query  $t$  at offset 0 and the gripper command from query  $t - 20$  at offset 20. A20G0 swaps these assignments: the arm comes from query  $t - 20$  at offset 20, and the gripper comes from query  $t$  at offset 0. During the first 20 steps, before the older source exists, every component executes the Fresh offset-0 command.

Formally, let  $q$  be policy-query time,  $k$  the chunk offset, and  $A_q[k]$  the predicted action entry. A valid same-target source satisfies

$$q + k = t. \quad (1)$$

The source age is  $d = t - q$ , so same-target alignment requires  $k = d$ . We therefore measure the joint effect of an older source observation and a longer-lookahead prediction; the probe does not separate those factors. Translation comprises action dimensions 0–2, rotation dimensions 3–5, and the gripper dimension 6. Same-target probes query the policy once per executed step, giving a policy-query rate of 1.0, and discard unselected predictions. They are diagnostic interventions, not deployment executors.

Executable schedules use separate notation. Periodic  $Hh$  queries every  $h$  steps and commits the full action to that chunk, giving the frozen H16, H4, and H2 schedules. ARM $h$ \_GRIP $g$  assigns separate commitment horizons to the arm and gripper, yielding ARM4\_GRIP32 and ARM2\_GRIP16. TE\_DENSE queries at every step and applies the canonical temporal ensemble. These frozen schedules test an operational consequence; they are not a new adaptive executor.

## IV. Experimental Setup

### A. Policy, clock, and action interface

We evaluate fixed ACT policies [1] using LeRobot v0.4.4 [17] in LIBERO simulation [2]. The evaluator runs at 20 Hz, so one chunk index is 0.05 s and  $d = 20$  is 1.00 s. ACT predicts 100-step chunks without action smoothing. Unless temporal ensembling is explicitly named, it is disabled. The policy outputs a seven-dimensional relative end-effector action with the component grouping defined in Sec. III.

### B. Cohorts, controls, and inference

A paired block is one task–initial-state pair evaluated under each compared condition with matched seeds, preprocessing, step caps, and success criteria. The primary preregistered cohort contains 140 blocks across five LIBERO-Goal and five LIBERO-10 tasks. The Object development cohort contains 126 blocks across nine

TABLE I

Reader-facing evidence provenance. Development and later controls characterize or bound the primary result; they are not pooled with it.

| Evidence                           | Role                                |
|------------------------------------|-------------------------------------|
| 140-block same-target test         | Preregistered primary evaluation    |
| Object age/component probes        | Development-cohort characterization |
| Dense-query and mechanism controls | Reviewer-directed controls          |
| Spatial factorial completion       | Later, separate scope comparison    |
| Track A                            | Preregistered executable schedules  |

tasks. Track A contains 450 blocks across LIBERO-10, Goal, and Spatial.

The dense-query control factorial uses C00 for Fresh arm and gripper, C10 for H16 arm and Fresh gripper, C01 for Fresh arm and H16 gripper, and C11 for H16 arm and gripper. Every condition queries the policy at each executed step and discards unused predictions. Success is binary. Paired contrasts use two-sided exact McNemar tests. We report paired risk-difference intervals and task-cluster bootstrap intervals already fixed in the canonical analysis.

Primary confirmatory conditions, cohorts, and decision rules were fixed before outcome inspection. Before selecting held-out confirmation cells, we audited the available experiment history for prior outcome exposure and excluded cells with existing outcomes. Development and reviewer-directed characterization are reported separately rather than pooled with the preregistered primary cohort.

## V. Results

### A. Q1: Same-Target Arm-Gripper Asymmetry

Moving the same one-second-old prediction from the gripper to the arm reduced success from 83/140 (59.29%) to 38/140 (27.14%), a 32.14-point difference (Fig. 1). Under the sign convention  $S(\text{A0G20}) - S(\text{A20G0})$ , discordance was 48:3 and the exact McNemar test gave  $p = 1.96749 \times 10^{-11}$ . The paired 95% interval was  $[+23.6, +40.7]$  points, and the task-cluster interval was  $[+21.4, +44.3]$  points. This preregistered diagonal contrast establishes that assigning the stale relationship to the arm or gripper is not behaviorally interchangeable, but heterogeneous conditional effects prevent a unique additive arm/gripper attribution.

The same asymmetry appeared when we later completed the previously missing Spatial comparison. Fresh-arm/stale-gripper execution succeeded on 40/100 blocks and stale-arm/fresh-gripper execution on 12/100, a +28.00-point difference with paired

TABLE II

Complete Object-126 same-target age sweep. SA/FG denotes stale arm/fresh gripper; FA/SG denotes fresh arm/stale gripper. Separations are FA/SG minus SA/FG in percentage points.

| $d$ | Time   | SA/FG           | FA/SG           | Separation |
|-----|--------|-----------------|-----------------|------------|
| 2   | 0.10 s | 49/126 (38.89%) | 76/126 (60.32%) | +21.43     |
| 4   | 0.20 s | 49/126 (38.89%) | 82/126 (65.08%) | +26.19     |
| 8   | 0.40 s | 36/126 (28.57%) | 88/126 (69.84%) | +41.27     |
| 12  | 0.60 s | 21/126 (16.67%) | 88/126 (69.84%) | +53.17     |
| 16  | 0.80 s | 23/126 (18.25%) | 90/126 (71.43%) | +53.17     |
| 20  | 1.00 s | 12/126 (9.52%)  | 81/126 (64.29%) | +54.76     |
| 32  | 1.60 s | 32/126 (25.40%) | 79/126 (62.70%) | +37.30     |

interval  $[+17.00, +39.00]$  and task-cluster interval  $[+13.00, +45.00]$ . This separate comparison preserves the sign but is not pooled into the original 140-block result.

### B. Q2: Temporal Scale and Component Identity

The fresh-arm/stale-gripper branch remained above the stale-arm/fresh-gripper branch at every tested age from 0.10 to 1.60 s on the Object development cohort (Fig. 2, left; Table II). Observed separations ranged from +21.43 to +54.76 points, with no ordering reversal and the largest values between 0.60 and 1.00 s. At 1.00 s, the separation was the largest observed on the grid, but no lag-versus-lag inference establishes a distinct peak.

Neither absolute branch was monotone. Stale-arm/fresh-gripper success was 38.89, 38.89, 28.57, 16.67, 18.25, 9.52, and 25.40% as age increased through 0.10, 0.20, 0.40, 0.60, 0.80, 1.00, and 1.60 s. It reached its observed minimum at 1.00 s and recovered at 1.60 s. Fresh-arm/stale-gripper success was 60.32, 65.08, 69.84, 69.84, 71.43, 64.29, and 62.70%. It was highest at 0.80 s and lower at 1.00 and 1.60 s. These shapes do not define a tolerance threshold for either component.

Translation staleness was substantially more damaging than rotation staleness on the same development cohort (Fig. 2, right). Translation-stale execution succeeded on 11/126 blocks (8.73%), whereas rotation-stale execution succeeded on 53/126 (42.06%). The primary contrast  $S(\text{translation stale}) - S(\text{rotation stale})$  was  $-33.33$  points, with discordance 3:45, exact  $p = 1.31259 \times 10^{-10}$ , paired interval  $[-42.06, -24.60]$ , and task-cluster interval  $[-44.44, -22.22]$  points. All frozen leave-one-task-out estimates were negative. This result supports component-dependent temporal sensitivity within the evaluated ACT policy on this cohort.

Rotation staleness incurred little detectable cost relative to Fresh on this cohort. Rotation-stale success was 42.06%, compared with 44.44% for Fresh, a  $-2.38$ -point contrast ( $p = 0.607$ ); the paired interval  $[-8.73, +3.97]$  and task-cluster interval  $[-6.35, +1.59]$  both included zero. The exactly matched 1.00-s characterization was 8.73% for translation stale, 9.52% for the full arm stale, 42.06% for rotation stale, 44.44% for Fresh, and 64.29% for gripper stale. Staling translation alone nearly reproduces the degradation observed when the full arm

Draft Fig. 1 specification. Left: a control-step timeline showing  $A_t[0]$  and  $A_{t-20}[20]$  converging on physical target  $t$ , with  $q + k = t$ ,  $d = 20 = 1.00$  s at 20 Hz, and the all-Fresh prefix for  $t < 20$ . Center: A0G20 uses fresh arm/stale gripper; A20G0 uses stale arm/fresh gripper. Right: the preregistered 140-block factorial, visually emphasizing only A0G20–A20G0=+32.14 points, paired CI [+23.6, +40.7], and task-cluster CI [+21.4, +44.3].

Fig. 1. Same-target diagnostic and primary preregistered result. At step  $t$ , A0G20 combines the arm from query  $t$ , offset 0, with the gripper from query  $t - 20$ , offset 20; A20G0 swaps those assignments. Both commands target step  $t$ . Moving the one-second-old prediction from the gripper to the arm changed success from 83/140 to 38/140, a 32.14-point difference. These probes query the policy once per executed step (policy-query rate 1.0) and are diagnostic interventions rather than deployment executors.

Draft Fig. 2 specification. Left: absolute Object-126 success curves for stale arm/fresh gripper and fresh arm/stale gripper at 0.10, 0.20, 0.40, 0.60, 0.80, 1.00, and 1.60 s. Mark 1.00 s as the originally frozen lag, not an optimum. Right: five block-matched 1.00-s values, with distinct encodings for historical anchors and later probes: translation stale 8.73%, full arm stale 9.52%, rotation stale 42.06%, Fresh 44.44%, and gripper stale 64.29%. Annotate translation-minus-rotation as -33.33 points.

Fig. 2. Age dependence and within-arm component structure on the Object development cohort. The fresh-arm/stale-gripper branch remains above the stale-arm/fresh-gripper branch at every tested age, with observed separations from +21.43 to +54.76 points; both curves are non-monotone. The 1.00-s separation is the largest observed on the grid, but no lag-versus-lag inference establishes a distinct peak. Historical anchors and reviewer-directed probes are exactly block matched but were generated at different study stages. The right panel is therefore a comparative characterization, not a single five-condition preregistered factorial. The stale-gripper effect magnitude should not be generalized across cohorts.

Draft Fig. 3 specification. Focal ordering contrast: same-target dispersion rotation > translation, but behavioral sensitivity translation > rotation. Secondary rows: cross-policy localization interval crossing zero; gripper persistence 0.675 at 1.00 s; occupancy  $\rho = 0.192$ ,  $p = 0.309$ ; forecastability descriptive. Omit command discontinuity from the main panel.

Fig. 3. Mechanism accounting. Same-target disagreement predicts the within-arm ordering incorrectly. Persistence, occupancy, forecastability, and command-discontinuity diagnostics do not identify a positive causal mechanism.

is stale. Historical anchors and later component probes were generated at different study stages, and the magnitude of the stale-gripper simple effect varies across cohorts.

### C. Q3: Do Simple Diagnostics Explain the Ordering?

Same-target prediction dispersion ranks the arm components in the opposite order from behavioral sensitivity (Fig. 3). Normalized ACT dispersion ranked rotation (0.1484) above translation (0.1364), whereas the behavioral intervention ranked translation staleness as much more damaging than rotation staleness. The disagreement measurement remains valid, but it fails as an ordering predictor within the arm.

The tested diagnostics do not identify a positive causal mechanism.

### D. Q4: Executable Consequences and Query-Matched Controls

Under identical dense querying, extending gripper commitment produced a clear gain when the arm was also committed. C11–C10 was +12.14 points, and both canonical intervals were above zero (Table IV). The corresponding C01–C00 change with a fresh arm was +2.86 points, with both intervals spanning zero. Absolute success was 77/140 (55.00%) for C00, 76/140 (54.29%) for C10, 81/140 (57.86%) for C01, and 93/140 (66.43%) for C11.

Query schedule therefore cannot by itself explain this conditional effect in the tested deterministic ACT evaluator. All four conditions queried the policy at every executed step and discarded unused predictions. In a preregistered identity check, dense C11 and sparse H16 produced exactly the same executed actions, episode length, completion step, and terminal outcome from the same initial state. This identity statement is limited to the tested deterministic evaluator.

Interaction estimates remain supporting evidence and are not pooled. The original 140-block post-study interaction was +15.71 points, with paired-block interval [+6.43, +25.00], task- $t$  interval [-2.76, +34.18], and exact task sign-flip  $p = 0.101562$ . The R1C interaction was fixed before its outcomes and was +9.29 points, but no canonical interaction-specific interval,  $p$ -value, or leave-one-out estimate exists. Spatial has no named canonical interaction estimate. These point estimates do not establish replicated interaction. The R1C estimate

TABLE III

Mechanism-diagnostic accounting. Each observation is retained at its measured scope; none identifies a positive causal explanation.

| Candidate signal          | Observation                                                                                                | What it tells us                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Same-target dispersion    | ACT ranks rotation > translation > gripper; behavior ranks translation staleness above rotation staleness. | The metric does not predict the within-arm behavioral ordering.                     |
| Cross-policy localization | The ACT-minus-SmolVLA localization interval is $[-0.0422, +0.2815]$ .                                      | Lower gripper dispersion is not ACT-specific.                                       |
| Gripper persistence       | $P(\text{no transition within 1.00 s}) = 0.675018$ .                                                       | Persistence alone cannot explain useful stale-gripper predictions over this window. |
| Occupancy moderator       | $n = 30$ , Spearman $\rho = 0.1922$ , $p = 0.3089$ .                                                       | The preregistered moderator detects no task-level association.                      |
| Command discontinuity     | Conditions differ in visited states, trajectories, and prediction offsets.                                 | The comparison cannot distinguish trajectory composition from execution smoothness. |
| B3 forecastability        | No discriminative curve pattern was fixed in advance; the curves did not match the behavioral ordering.    | The curves are descriptive and do not distinguish the observed ordering.            |

TABLE IV

Dense-query RIC contrasts. Every condition queries the policy at each executed step. Effects use first condition minus second condition; discordance is first-only:second-only success. Intervals are in percentage points.

| Contrast | Effect (pp) | Discordance | Exact $p$   | Paired 95% CI     | Task-cluster 95% CI |
|----------|-------------|-------------|-------------|-------------------|---------------------|
| C10-C00  | -0.71       | 18:19       | 1           | $[-9.29, +7.86]$  | $[-12.14, +10.71]$  |
| C11-C01  | +8.57       | 26:14       | 0.0806905   | $[+0.00, +17.14]$ | $[-2.14, +21.43]$   |
| C01-C00  | +2.86       | 10:6        | 0.454498    | $[-2.86, +8.57]$  | $[-4.29, +9.29]$    |
| C11-C10  | +12.14      | 20:3        | 0.000488281 | $[+5.71, +18.57]$ | $[+3.57, +21.43]$   |
| C11-C00  | +11.43      | 28:12       | 0.016589    | $[+2.86, +20.00]$ | $[-0.71, +23.57]$   |
| C10-C01  | -3.57       | 12:17       | 0.458258    | $[-11.43, +4.29]$ | $[-14.29, +6.43]$   |

Draft Fig. 4 specification. Main: pooled success versus log policy queries per executed environment step for H16, H4, ARM4\_GRIP32, H2, ARM2\_GRIP16, and TE\_DENSE. Use points only, no regression or connecting line; give TE\_DENSE a distinct marker. Annotate H4→ARM4\_GRIP32 (+4.667 points) and H2→ARM2\_GRIP16 (+5.778 points). Identify H16 as strongest overall at 79.33%. Inset: both paired effects for LIBERO-10, Goal, and Spatial, if legible at column width.

Fig. 4. Execution consequence at constrained replanning cadence. Extending gripper commitment improves success at two fixed arm cadences and nearly matched policy-query rates per executed step. Coherent H16 remains the strongest overall frozen operating point. The paired gains concentrate in LIBERO-10; suite identity, baseline difficulty, and task semantics covary, so the source of heterogeneity is not identified. Policy-query rate is not equal compute.

used matched policy-query schedules, so the observed conditional pattern is not simply created by one condition querying more often.

Component-resolved commitment also improved executable schedules at two fixed arm cadences (Fig. 4). Extending only gripper commitment from H4 to ARM4\_GRIP32 increased success from 314/450 to 335/450, or +4.667 points (paired interval  $[+2.00, +7.56]$ ; task-cluster interval  $[+0.67, +9.11]$ ). At the faster cadence, H2 to ARM2\_GRIP16 increased success from 295/450 to 321/450, or +5.778 points

(paired interval  $[+3.56, +8.22]$ ; task-cluster interval  $[+2.67, +9.56]$ ). Policy-query rates per executed step were nearly identical within each pair, approximately 0.251 and 0.501, respectively.

Coherent H16 nevertheless remained the strongest overall frozen operating point at 357/450 successes (79.33%). Component-resolved schedules therefore have an operational consequence under constrained replanning cadence but are not globally superior executors. Their gains were concentrated in LIBERO-10: the H4 pair improved by 13.333 points there, 0.000 on Goal, and 0.667 on Spatial; the H2 pair improved by 14.000, 2.000, and 1.333 points. H16 success was 54.7% on LIBERO-10, 91.3% on Goal, and 92.0% on Spatial. Suite identity, baseline difficulty, and task semantics covary, so these data do not identify the source of the concentration.

TE\_DENSE was a secondary frozen Track-A condition, not a headline method comparison. It achieved 288/450 successes (64.00%), compared with 357/450 (79.33%) for H16. Under the TE\_DENSE-minus-H16 convention, the difference was -15.333 points, discordance 14:83, exact  $p = 4.22241 \times 10^{-13}$ , paired interval  $[-19.333, -11.333]$ , and task-cluster interval  $[-21.556, -10.000]$  points.

Under this frozen canonical configuration, dense temporal aggregation placed substantial weight on old predictions and produced many more near-boundary gripper commands. TE\_DENSE used LeRobot v0.4.4 ACTTemporalEnsembler with coefficient 0.01 and chunk length

TABLE V

Frozen Track-A execution results. Query rate is measured policy queries per executed environment step. Suite entries report successes/150 and percentage.

| Condition   | Pooled success   | Query rate | LIBERO-10      | Goal            | Spatial         |
|-------------|------------------|------------|----------------|-----------------|-----------------|
| H16         | 357/450 (79.33%) | 0.064864   | 82/150 (54.7%) | 137/150 (91.3%) | 138/150 (92.0%) |
| H4          | 314/450 (69.78%) | 0.251156   | 61/150 (40.7%) | 125/150 (83.3%) | 128/150 (85.3%) |
| ARM4_GRIP32 | 335/450 (74.44%) | 0.251318   | 81/150 (54.0%) | 125/150 (83.3%) | 129/150 (86.0%) |
| H2          | 295/450 (65.56%) | 0.500720   | 53/150 (35.3%) | 117/150 (78.0%) | 125/150 (83.3%) |
| ARM2_GRIP16 | 321/450 (71.33%) | 0.500907   | 74/150 (49.3%) | 120/150 (80.0%) | 127/150 (84.7%) |
| TE_DENSE    | 288/450 (64.00%) | 1.000000   | 55/150 (36.7%) | 120/150 (80.0%) | 113/150 (75.3%) |

100. The upstream ordering gives greater weight to older predictions for a positive coefficient, and a runtime check confirmed that direction. Its empirical weighted mean prediction age was 44.99 steps (2.249 s), with a weighted median of 2.15 s and 95th percentile of 4.70 s; 52.27% of normalized weight fell on predictions older than 2.0 s. The native gripper command satisfied  $|g| < 0.50$  on 24.41% of steps, while its 1.02% switch rate was the lowest among the frozen executors. These observations characterize the configuration but do not establish that either age weighting or intermediate gripper commands caused its performance loss.

## VI. Discussion and Limitations

A single temporal execution rule is not behaviorally neutral across the components of the evaluated ACT actions. The preregistered same-target contrast shows a large arm–gripper asymmetry, the development sweep preserves its ordering over 0.10–1.60 s, and the within-arm comparison separates translation from rotation. The practical consequence is narrower: longer gripper commitment improves two schedules when arm cadence is fixed, but coherent H16 remains the strongest overall frozen operating point.

Same-target alignment preserves the physical step for which every command was predicted, but it couples two temporal quantities. An older source observation necessarily contributes a farther-lookahead prediction because  $k = d$ . The diagnostic therefore estimates their joint closed-loop effect and cannot say whether source age, prediction horizon, or their combination produces the behavioral difference. This limitation is part of the intervention definition, not merely a statistical qualification.

The age sweep and translation–rotation comparison characterize an Object development cohort rather than the primary confirmatory cohort. Their blocks are exactly matched across the conditions being compared, but historical anchors and later reviewer-directed probes were generated at different study stages. The separate Spatial completion preserves the main asymmetry’s sign, yet it was also conducted after the primary study and was not pooled with the 140-block result.

Same-target probes are diagnostic interventions, not candidate deployment schedules. They query the policy

at every executed step even when an older prediction supplies part of the action. The dense-query factorial shows that discarded queries alone cannot explain one conditional H16 effect under the tested deterministic evaluator. Track A then supplies the deployment-facing comparison at fixed arm cadences and nearly matched policy-query rates. Query rate is not a measurement of equal latency, total query count, or equal compute.

The Track-A gains are heterogeneous across suites. Both component-resolved contrasts are largest in LIBERO-10, while Goal and Spatial operate near the H16 ceiling. Because suite identity, baseline difficulty, and task semantics vary together, their individual roles cannot be identified. This heterogeneity and the 357/450 H16 result rule out a claim that component-resolved scheduling is a globally superior executor in the evaluated static benchmark.

The tested diagnostics do not identify a positive mechanism. Same-target dispersion predicts the wrong within-arm ordering, gripper persistence is incomplete, the occupancy moderator is null, forecastability lacks a prespecified discriminative curve pattern, and command discontinuity cannot separate the competing explanations. These results narrow the explanation space without resolving why translation, rotation, and gripper commands respond differently.

Finally, the central behavioral evidence is specific to the evaluated ACT policies, LIBERO tasks, checkpoints, and control interface. SmolVLA [18] cannot provide a physically matched replication because its training and chunk physical timebase cannot be identified from the available provenance. A planned SmolVLA evaluation was not run because its prespecified runtime eligibility window had expired. We therefore make no cross-policy or physical-robot claim. Broader action spaces, stochastic evaluators, and real robots remain outside the evidence reported here.

## VII. Conclusion

A single temporal execution rule is not behaviorally neutral across ACT action components. The ordering persists over a broad range of source ages and extends to translation versus rotation within the arm, while simple same-target disagreement ranks those components incorrectly. Component-resolved gripper commitment can

improve execution at fixed arm replanning cadence, but coherent H16 remains the strongest overall operating point. The mechanism and generality beyond the evaluated ACT policies remain open.

## References

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